# LAB 5

## Introduction to **Object-Oriented Programming**

## Data Storage

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# Lab05 Exercises

## ❖ Lab05.1. Date

Write a program to allow user to work with date and time. Create a class called “MyDate” with following methods:

- getCurrentDate
- calculateDaysBtwDates
- findDay

***You can use any libraries in Java related to Date and Time. Ex. **Date, SimpleDateFormat, or Calendar**

**Console:**

```
==== Menu ===
1.  Current date and time
2.  Calculate days btw two dates
3.  Find the day of the week
4.  Quit
Choose an opt: 1
Current datetime is: 02-1-2023 06:07:59

Choose an opt: 2
First date (dd/mm/yyyy): 02/02/2023
Second date (dd/mm/yyyy): 13/02/2023
Difference between two dates is: 10 days

Choose an opt: 3
Input a date: 02/02/2023
The day is: Wednesday
```

Give me the clue please???

```java
SimpleDateFormat format = new SimpleDateFormat("dd-MM-yyyy HH:mm:ss");
Date d = format.parse("01/02/2023");
long milliseconds = d.getTime();
```

```java
SimpleDateFormat format = new SimpleDateFormat("dd/MM/yyyy");
Date date = format.parse("01/02/2023");
Calendar calendar = Calendar.getInstance();
calendar.setTime(date);
int dayOfWeek = calendar.get(Calendar.DAY_OF_WEEK);
```

# Lab05 Exercises

## ❖ Lab05.2. Multi deletion

Write a program to allow user to *infinitely input student into* a table and delete many at once.
You can suggest Attributes (id, name, age, etc.)

**Console:**

```
==== Menu ===
1.   Add new students
2.   Delete multiple students
3.   Quit
Choose an opt: 1

Student #1:
Id: 111
Name: Tola
Age: 17

Do you want to add more (y/n)?: y
Student #2:
Id: 222
Name: Makara
Age: 16

Do you want to add more (y/n)?: y
Student #3:
Id: 333
Name: Kompheak
Age: 25

Do you want to add more (y/n)?: n
```

**Console:**

```
==== Menu ===
1.   Add new students
2.   Delete multiple students
3.   Quit
Choose an opt: 2

=============================================================
| No        | ID        | Name                  | Age       |
=============================================================
| 1         | 111       | Tola                  | 17        |
| 2         | 222       | Makara                | 16        |
| 3         | 333       | Kompheak              | 25        |
=============================================================
==== DELETION =====
Input student #1 ID: 111
Do you want to delete more (y/n)?: y
Input student #2 ID: 444
Do you want to delete more (y/n)?: y
Input student #3 ID: 222
Do you want to delete more (y/n)?: n

Student with ID 111 deleted
Student with ID 444 not found!
Student with ID 222 deleted
```

# Lab05 Exercises

## ❖ Lab05.3. Shape mamagement (apply inheritance concept to create classes of shape)

Write a program to allow user mange different shapes of geometry.

You can suggest Attributes according to the shapes Line, Rectangles and Triangle

**Console:**

```
==== Menu ===
1.  View all lines
2.  View all rectangles
3.  View all triangles
4.  Add a new shape
5.  Quit
Choose an opt: 3

==== Add new shape ====
Select a shape:
1.  Line
2.  Rectangle
3.  Triangle
Choose an opt: 1
Input length: 5

==== Add new shape ====
Select a shape:
1.  Line
2.  Rectangle
3.  Triangle
Choose an opt: 2
Input width: 5
Input height: 3
```

**Console:**

```
==== Menu ===
1.  View all lines
2.  View all rectangles
3.  View all triangles
4.  Add a new shape
5.  Quit
Choose an opt: 1

==== List all lines ====
1. length: 5
_____
2. length: 30
____________________________
3. length: 10
__________

Choose an opt: 2
1. Width: 5, Height: 3
*****
*   *
*****

2. Width: 3, Height: 3
***
* *
***
```

# Lab05 Exercises

## ❖ Lab05.4. Text Editor

Write a program to record your note

→ Data must be permanently stored in local file (Ex. data.txt)

**Console:**

```
==== Menu ===
1.  View my note
2.  Edit
3.  Quit
Choose an option: 2

==== Edit following note ====
****************************
1| Everybody look to their left
2| Everybody look to their right
3| Can you feel that? Yeah
4| We're paying with love tonight
****************************
1.  Append new line
2.  Update at line
3.  Delete line
Choose an opt: 1
Input a string for line #5:
It's not about the money, money, money
Line #5 is appended to the note.

Choose an opt: 3
Delete line number: 2
Line #2 is deleted.
```

**Console:**

```
==== Menu ===
1.  View my note
2.  Edit
3.  Quit
Choose an option: 1

==== View note ====
****************************
1| Everybody look to their left
2| Can you feel that? Yeah
3| We're paying with love tonight
4| It's not about the money, money, money
****************************
```

# Lab05 Exercises

## ❖ Lab05.5. Human resource in a school

Write a program to manage human resource of school

→ User data must be permanently stored in local file (Ex. teacher.txt, student.txt, securityguard.txt)

**Console:**

```
==== Menu ===
1.  View all
2.  Add new
3.  Quit
Choose an option: 2

==== Add new resource ====
1. Teacher
2. Student
3. Security guard
Choose an opt: 1
First Name: Tola
Last Name: Ma
Sex: M
Email: tola@domain.com
Subject: OOP, C++, Web dev
Salary: 100

Choose an opt: 1
First Name: Makara
Last Name: John
Sex: F
Email: makara@domain.com
Year: 2
Major: Computer science
```

**Console:**

```
==== Menu ===
1.  View all
2.  Add new
3.  Quit
Choose an option: 1

==== Teacher ====
1. [Tola Ma][M][tola@domain.com][OOP, C++, Web dev][100$]
2. [Tola Ma][M][tola@domain.com][OOP, C++, Web dev][100$]
3. [Tola Ma][M][tola@domain.com][OOP, C++, Web dev][100$]

=========== Student ===========
1. [Jonh Makara][F][makara@domain.com][2][Computer science]
2. [Jonh Makara][F][makara@domain.com][2][Computer science]

=========== Security guard ===========
1. [Peter Pronh][M][pronh@domain.com][door guard]
2. [Peter Pronh][M][pronh@domain.com][security]
```

# Lab05 Exercises

## ❖ Lab05.6. Authentication

Write a program to manage authentication of a system.

→ User data must be permanently stored in local files (Ex. user.txt)

**Console:**

```
==== Menu ===
1.  Login
2.  Register
3.  Quit
Choose an option: 2

==== Register a new user ====
First Name: E
Last Name: No
Username: eno
Email: eno@domain.com
Password: 12345

You are successfully registered~
```

**Console:**

```
==== Menu ===
1.  Login
2.  Register
3.  Quit
Choose an option: 1

==== Login ====
Email or Username: eno
Password: 123456
User not found. Please try again

Email or Username: eno
Password: 12345
=========== User Info ===========
Hi E No,
Your username is: eno
Your email is: eno@domain.com
```

# Good luck 👌

## ✋ How to submit my Lab works???

👉 Screenshot your source codes and answers as a single **pdf** file and submit to the **Moodle** with Java file attachment as well.

**Note: Don’t zip the files for submission**